NrgOpt Photovoltaic Project Site Survey Checklist

Project Information

Item	Detail
Project / Owner Name	
Project Address	
GPS Coordinates (lat/lng)	
Owner Contact Person	
Owner Contact Phone	
Survey Date	
Survey Engineer	
Weather During Survey	

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1. Building Specifications

1.1 Basic Dimensions

- [ ] Building area (m^2): _
- [ ] Elevation (m): _
- [ ] Number of floors: _
- [ ] Building orientation (sun-facing direction): _
- [ ] Year built / time in use: _
- [ ] Site usage / ownership type: _

1.2 Structural Load

- [ ] Roof design live load (kg/m^2): _
- [ ] Current load status (existing equipment weight): _
- [ ] Remaining capacity (kg/m^2): _
- [ ] Structural reinforcement required? [ ] Yes [ ] No
- [ ] Building seismic rating: _
- [ ] Structural inspection report available? [ ] Yes [ ] No

1.3 Roof Type

Concrete Roof:

- [ ] Length x Width x Height (m): _
- [ ] Parapet wall height (m): _
- [ ] Staircase / opening locations and dimensions: _
- [ ] Roof pitch / slope: _
- [ ] Water leakage condition (describe): _
- [ ] Any repairs, contamination, or equipment on roof? _
- [ ] Skylight area / light strips: _
- [ ] Obstacle dimensions (LxWxH): _
- [ ] Roof waterproofing condition: _

Metal Sheet Roof:

- [ ] Sheet profile type: [ ] Standing seam [ ] Angle brace [ ] Trapezoidal
- [ ] Roof pitch: _
- [ ] Beam specification: _
- [ ] Beam span (m): _
- [ ] Purlin specification: _
- [ ] Purlin span (m): _
- [ ] Column specification: _
- [ ] Purlin rust condition: _
- [ ] Roof surface rust condition: _
- [ ] Roof panel thickness (mm): _
- [ ] Does drawing match actual roof? [ ] Yes [ ] No

1.4 Obstruction Survey

- [ ] Obstruction 1: L__ **x W_ *x H (m), Direction from roof centre:* ____**
- [ ] Obstruction 2: L__ **x W_ *x H (m), Direction from roof centre:* ____**
- [ ] Obstruction 3: L__ **x W_ *x H (m), Direction from roof centre:* ____**
- [ ] Additional notes: ____**______**

1.5 Shading Analysis

- [ ] Surrounding buildings (height + distance from site):
- Building 1: height__**m, distance_*m, direction*___**
- Building 2: height__**m, distance_*m, direction*___**
- [ ] Trees or vegetation casting shadows: [ ] Yes [ ] No
- [ ] Chimneys / exhaust pipes nearby: [ ] Yes [ ] No
- [ ] Shading from adjacent roof structures: [ ] Yes [ ] No
- [ ] Winter solstice sun path checked? [ ] Yes [ ] No
- [ ] Drone shadow simulation performed? [ ] Yes [ ] No

1.6 Solar Panel Layout

- [ ] Rail spacing (mm): _
- [ ] Tilt angle planned (°): _
- [ ] Row spacing (m): _
- [ ] Panel orientation: [ ] Landscape [ ] Portrait
- [ ] Estimated number of panels: _
- [ ] PV power station height (m): _

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2. Meteorological & Environmental

2.1 Climate Data

- [ ] Annual solar irradiation (kWh/m²): _
- [ ] Maximum wind speed (m/s): _
- [ ] Snow load (kN/m²): _
- [ ] Extreme low temperature (°C): _
- [ ] Extreme high temperature (°C): _
- [ ] Annual average temperature (°C): _
- [ ] Annual rainfall (mm): _
- [ ] Lightning frequency: [ ] Low [ ] Medium [ ] High

2.2 Environmental Risks

- [ ] Within 5km of coastline? [ ] Yes [ ] No
- [ ] Nearby chemical / cement / steel plants? [ ] Yes [ ] No
- [ ] Nearby water bodies (flood risk)? [ ] Yes [ ] No
- [ ] Earthquake-prone zone? [ ] Yes [ ] No
- [ ] Typhoon path? [ ] Yes [ ] No
- [ ] Soil corrosion level: [ ] Low [ ] Medium [ ] High
- [ ] Bird activity level: [ ] Low [ ] Medium [ ] High

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3. Electrical Infrastructure

3.1 Grid Connection

- [ ] Grid connection type: [ ] Low-voltage [ ] High-voltage
- [ ] Number of transformers: _
- [ ] Transformer capacity (kVA): _
- [ ] Transformer model / winding group: _
- [ ] Existing protection devices (model): _
- [ ] Existing solar PV system? [ ] Yes (capacity: ______kW) [ ] No
- [ ] Distance to grid connection point (m): _

3.2 Distribution Room

- [ ] Location / photo taken: [ ] Yes [ ] No
- [ ] Space for PV connection cabinet? [ ] Yes [ ] No
- [ ] Spare breaker slots available? [ ] Yes (how many: ____) [ ] No
- [ ] Cable trench available? [ ] Yes [ ] No
- [ ] AC cable requires road crossing? [ ] Yes [ ] No
- [ ] Single-line diagram available? [ ] Yes (attached) [ ] No

3.3 Consumption Profile

- [ ] Power generation mode: [ ] Full grid export [ ] Self-consumption + surplus export
- [ ] Contracted demand (kW): _
- [ ] Hourly load profile available? [ ] Yes [ ] No
- [ ] Peak load period: _:00 -- _:00
- [ ] Weekend / holiday operation? [ ] Yes [ ] No
- [ ] Night shift operation? [ ] Yes [ ] No

3.4 Monthly Consumption

Month	kWh	Month	kWh
Jan		Jul	
Feb		Aug	
Mar		Sep	
Apr		Oct	
May		Nov	
Jun		Dec	

Annual total: _ kWh

3.5 Electricity Tariff

- [ ] Electricity supply company: _
- [ ] Tariff category: [ ] Industrial [ ] Commercial [ ] Other
- [ ] Peak period rate (CNY/kWh): _
- [ ] Flat period rate (CNY/kWh): _
- [ ] Off-peak period rate (CNY/kWh): _
- [ ] Demand charge (CNY/kW/month): _

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4. Site Access & Logistics

4.1 Access Assessment

- [ ] Road width to site entrance (m): _
- [ ] Truck access (max vehicle length): _
- [ ] Crane working radius (m): _
- [ ] Crane parking position identified? [ ] Yes [ ] No
- [ ] Road closure / traffic permit required? [ ] Yes [ ] No
- [ ] Material staging area available? [ ] Yes (size: ______m²) [ ] No
- [ ] Elevator / hoist for material lifting? [ ] Yes [ ] No
- [ ] Construction power supply available? [ ] Yes [ ] No

4.2 Safety Considerations

- [ ] Roof edge protection / guardrails present? [ ] Yes [ ] No
- [ ] Roof access (stairs / ladder / hatch): _
- [ ] Fall protection anchor points? [ ] Yes [ ] No
- [ ] Emergency evacuation route identified? [ ] Yes [ ] No
- [ ] Asbestos or hazardous materials? [ ] Yes [ ] No

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5. Regulatory & Documentation

5.1 Owner Documentation

- [ ] Property ownership certificate
- [ ] Land use certificate
- [ ] Building construction drawings
- [ ] Structural calculation report
- [ ] Electrical single-line diagram
- [ ] Fire safety approval
- [ ] Environmental impact assessment
- [ ] Business license
- [ ] CAD files / PDF documents

5.2 Regulatory Check

- [ ] Local distributed PV filing requirements: _
- [ ] Grid company pre-approval required? [ ] Yes [ ] No
- [ ] Building height restriction applicable? [ ] Yes [ ] No
- [ ] Heritage / conservation area? [ ] Yes [ ] No
- [ ] Aviation restriction applicable? [ ] Yes [ ] No

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6. Photo Checklist

#	Shot Description	Taken
1	North-facing view from roof	
2	East-facing view from roof	
3	South-facing view from roof	
4	West-facing view from roof	
5	Roof panorama (drone / wide angle)	
6	Roof surface close-up x 4 corners	
7	Obstructions close-up (each)	
8	Parapet wall / roof edge detail	
9	Existing equipment on roof	
10	Distribution room -- exterior	
11	Distribution room -- interior	
12	Spare breaker slots	
13	Transformer nameplate	
14	Cable route roof-to-distribution room	
15	Site entrance and access road	
16	Surrounding buildings (ground level)	
17	Roof access point	
18	Drone video (full building)	
19	GPS-tagged location photo	
20	Team photo at site	

7. Cost & Timeline Estimation

- [ ] Estimated cable length (m): _
- [ ] Estimated cable trenching (m): _
- [ ] Special access equipment? [ ] Yes (specify: _) [ ] No
- [ ] Grid outage window available? [ ] Yes (dates: _) [ ] No
- [ ] Expected construction start: _
- [ ] Expected construction duration (days): _
- [ ] Any seasonal constraint? _

8. Sign-off

Role	Name	Signature	Date
Survey Engineer			
Project Manager			
Owner Representative			
Electrical Engineer			
Structural Engineer			

9. Additional Notes

NrgOpt -- Enterprise Clean Energy Solutions

Document Version: v2.0 | Last Updated: 2026-05-22

Complete this checklist before proceeding to technical proposal stage.